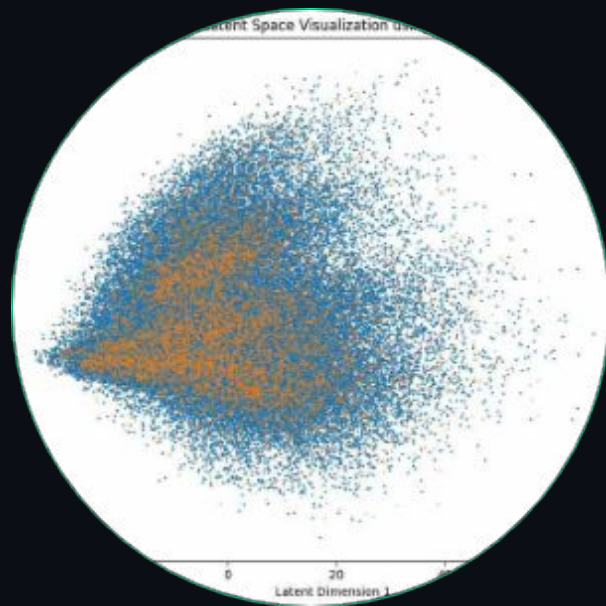




Generative AI: the state of the art

FP, FP and AP

Probabilistic Foundations

Generative models aim to learn the **joint probability distribution** of data, whereas discriminative models learn conditional boundaries.

$$P_{\text{model}}(X, Y) = P(X|Y)P(Y)$$

✓ **Sampling:** Drawing from the learned manifold. $X \sim P_{\text{data}}$

✓ **Density Estimation:** Explicitly modeling the likelihood.

Discriminative vs Generative

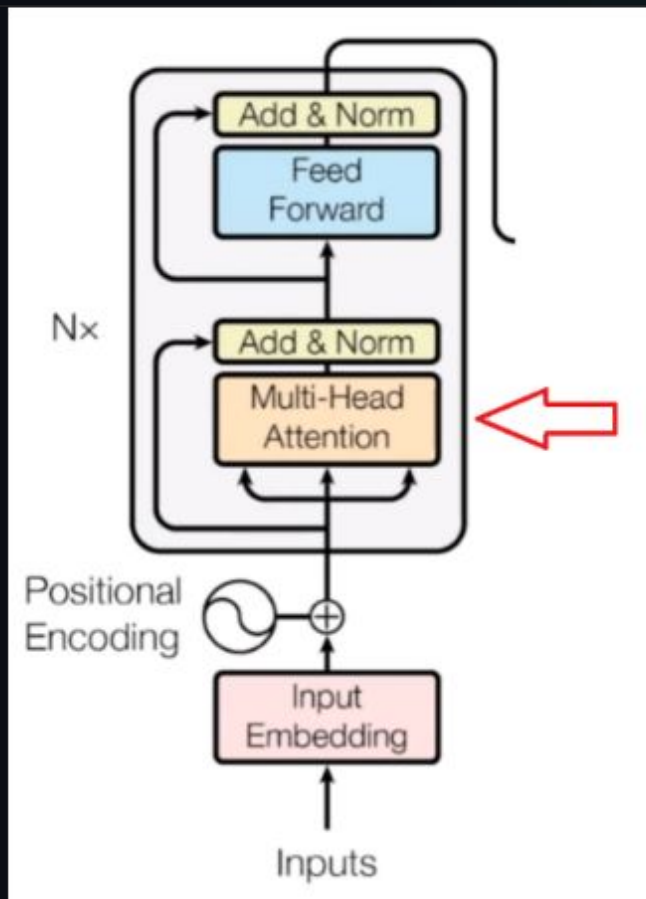
Discriminative: Maps high-dimensional input to a discrete label Y . Optimizes $P(Y|X)$

Generative: Models how data was generated. Can generate new X given (conditional) or just (unconditional).

| The Transformer Architecture

The paradigm shift from recurrent to **parallelizable attention** mechanisms (Vaswani et al., 2017).

- **No Recurrence:** Solves vanishing gradient in long sequences.
- **Positional Encoding:** Injects sequential order via sinusoidal functions.
- **Architecture:** Stacked Encoder (NLU) and Decoder (NLG) blocks.



| Attention Mechanism: The Core

Mapping Query, Key, and Value vectors to compute dynamic weightings.

$$\text{Attention} (Q , K , V) = \text{softmax} \frac{QK^T}{\sqrt{d_k}} V$$



Query (Q)

The vector representing the token currently seeking context from others.



Key (K)

The "index" vector used to calculate relevance (similarity) with the query.



Value (V)

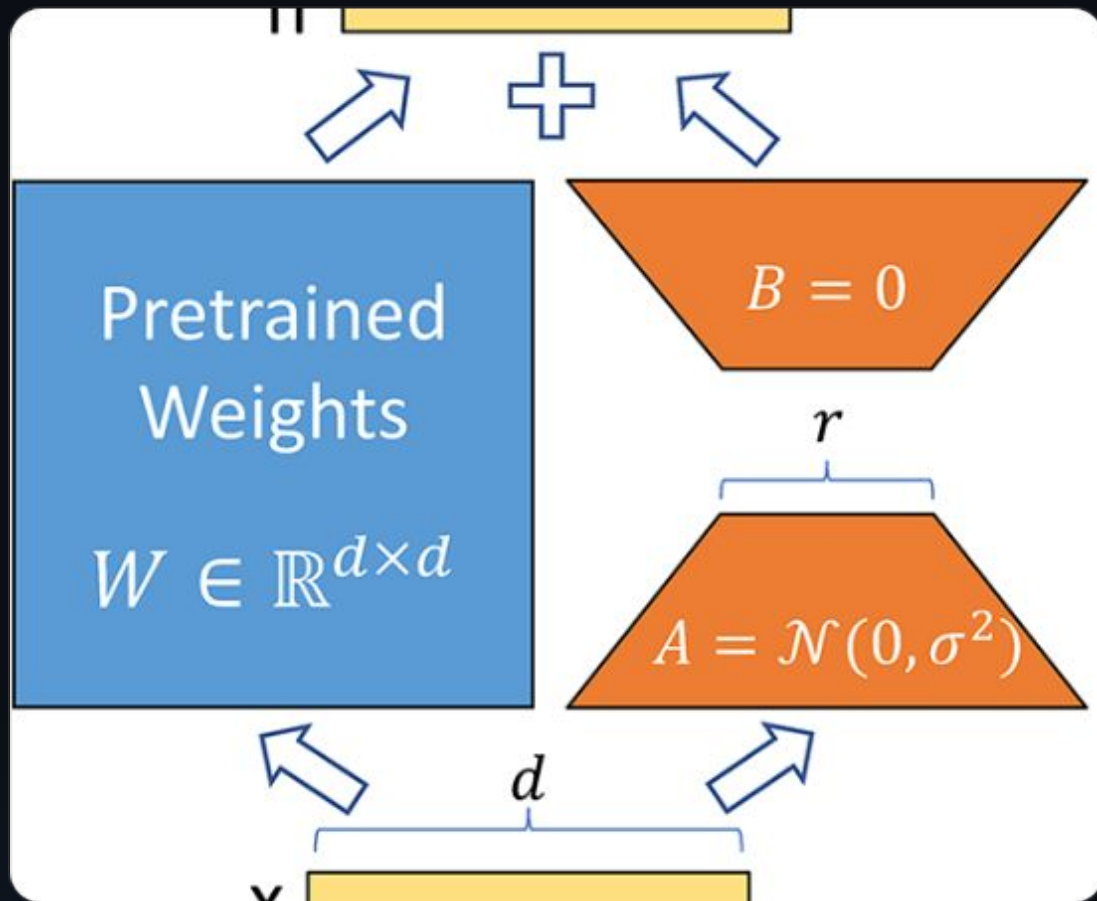
The content vector that is aggregated based on the attention weights.

| Stages of LLM Development



Scaling Laws: Performance follows a power law relative to Compute (C), Data (D), and Parameters (N).

Parameter-Efficient Fine-Tuning



LoRA (Low-Rank Adaptation) freezes the original weights and injects trainable rank-decomposition matrices.

$$h = W_0 x + \Delta W x = W_0 x + B A x$$

- Reduces trainable parameters by 10,000x.
- Minimal VRAM overhead (QLoRA: 4-bit).
- No inference latency (weights are merged).

Part II: Vision & Diffusion

Beyond Autoregressive Modeling

Diffusion Mechanics

Diffusion models generate data by reversing a Markov chain that slowly adds Gaussian noise to the input.

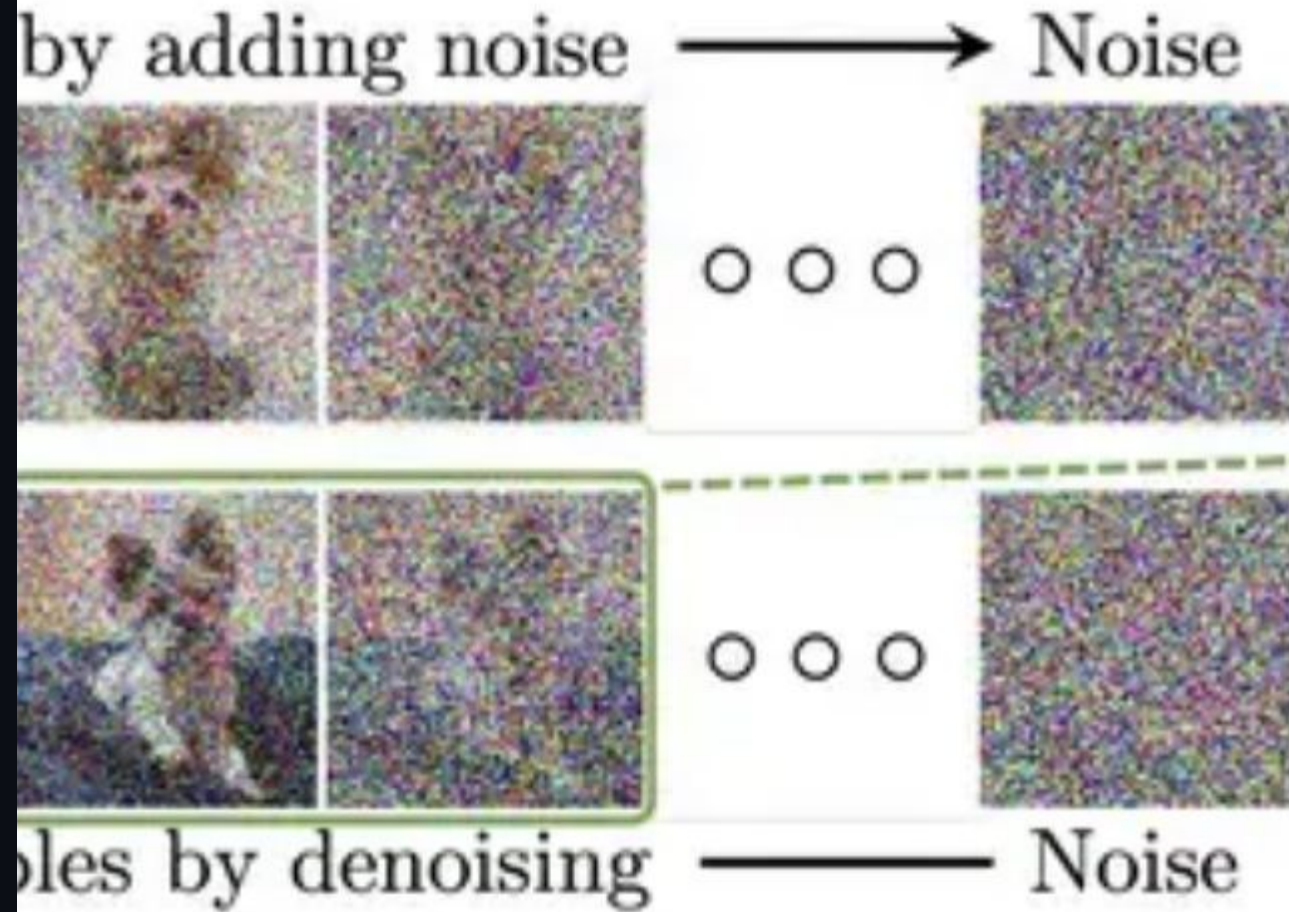
Key Components:

→ **Forward Process:** Adding noise until .

$$q(x_T) \approx \mathcal{N}(0, I)$$

↻ **Reverse Process:** Learning a U-Net to predict and subtract noise.

Optimized via **Denoising Score Matching**.



data by adding noise, then reverse this process. The reverse process requires estimating the score function (see [1]) with higher likelihood and less noise.

Variational Autoencoders (VAEs)

Learning a probabilistic latent space by maximizing the **Evidence Lower Bound (ELBO)**.

$$\text{ELBO} = \mathbb{E}_{q_{\psi}(z|x)} [\log p_{\theta}(x|z)] - D_{\text{KL}}(q_{\psi}(z|x) \parallel p(z))$$

The KL-divergence term acts as a regularizer, forcing the latent distribution to stay near a standard Normal prior.

Reparameterization Trick

To enable backpropagation through stochastic nodes:

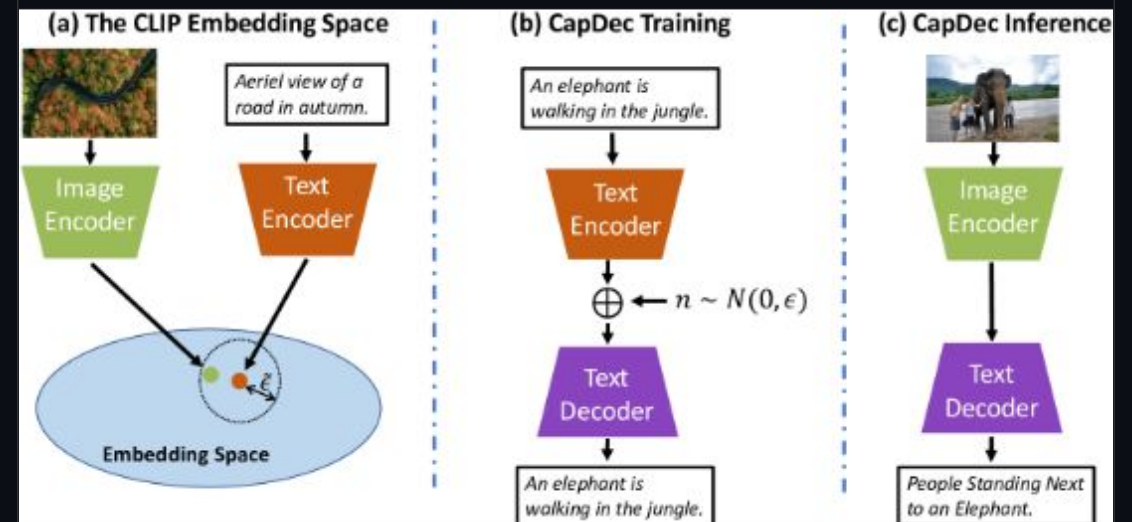
$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$$

Multimodal Alignment

CLIP (Radford et al., 2021)

Contrastive Language-Image Pre-training aligns images and captions in a shared embedding space.

- **Contrastive Loss:** Maximize cosine similarity for N correct pairs, minimize for N2-N wrong pairs.
- **Foundation:** Enables zero-shot classification and Stable Diffusion prompt guiding.



| Research Frontiers



Interpretability

Mechanistic interpretability: reverse-engineering weights into human-understandable circuits.



Long Context

Ring Attention and Linear Attention variants to process millions of tokens ($O(N)$ complexity).



Safety & Jailbreaking

Adversarial robustness and constitutional AI for alignment and refusal mechanics.

Next Step: Agentic Workflows and Tool-use (ReAct/Toolformer).